DEMYSTIFYING SENSORLESS FIELD-ORIENTED MOTOR CONTROL & FAULT-TOLERANT OBSERVERS

A Complete Layman's and Engineer's Guide to Motor Physics, Coordinate Mathematics, Embedded Firmware, and 13 Fault-Isolation Scenarios

Repository: sensorless-foc-pio | **Target MCU:** ESP32 Dual-Core (Xtensa 240 MHz) & Python/C SIL
Control Frequency: 20.0 kHz (50 µs loop tick) | **Observer:** Sliding Mode (SMO) + Flux Observer
Validation Status: 128 / 128 Automated Unit, Scenario & Robustness Tests PASSED (100% Green)
Author / System: Advanced Agentic Coding Architecture | Google DeepMind Pairing Protocol
Date: October 2026 | **Classification:** Educational & Technical Reference Manual

EXECUTIVE SUMMARY

The Big Picture: *Electric vehicles, surgical robots, multi-rotor drones, and high-speed trains demand electric motors that are silent, ultra-efficient, and rock-solid reliable. Historically, motors relied on physical position sensors (Hall-effect chips, optical encoders) attached to the spinning shaft. But physical sensors are fragile: their wires snap, vibration cracks their optics, and moisture destroys them. This repository provides a complete, production-grade* ***sensorless motor controller****: it calculates the rotor's exact physical angle and speed using pure mathematics (the motor's Back-EMF). Even more remarkably, it includes an advanced* ***fault-tolerant observer*** *that monitors the motor's current sensors using fundamental physical laws (Kirchhoff's Current Law). If a current sensor breaks, drifts, or suffers a loose connection, the controller isolates the liar within 15 ms and reconstructs the missing phase current algebraically (2-shunt mode) so the motor keeps spinning without a single hitch!*

Chapter	Section / Topic	Core Concepts & Physics Covered
1	**The Physics of Electric Motors**	Electromagnetism, Stator vs Rotor, 90° Torque Angle, Back-EMF, BLDC vs PMSM.
2	**Field-Oriented Control (FOC)**	The 3-phase AC dilemma, Clarke & Park transforms, d/q axis, PI current loops, SVPWM.
3	**The Virtual Sensor (Observers)**	Sliding Mode Observer (SMO), Flux Observer, Back-EMF tracking, PLL angle extraction.
4	**Inverter Realities & Power Electronics**	MOSFET half-bridges, shoot-through, dead-time distortion, body diodes, bus dynamics.
5	**Fault Detection & Fault Tolerance**	Kirchhoff's Current Law (KCL), hypothesis testing, 2-shunt reconstruction, debounce envelope.
6	**The 13 Test Scenarios in Detail**	Complete walkthrough of all 13 fault scenarios, acceptance thresholds, and physics.
7	**ESP32 Microcontroller Implementation**	Dual-core FreeRTOS, 20 kHz ISR, direct register ADC, cycle-counter benchmark.
8	**Glossary & Technical Reference**	Plain-English definitions for every acronym and parameter specification table.

Chapter 1: The Intuitive Physics of Electric Motors

To appreciate how our code controls an electric motor, we must first look at the physical principles of electromagnetism. At its core, an electric motor is a machine that converts electrical energy into mechanical twisting force (torque). It achieves this by harnessing the fundamental law of magnetism: **opposites attract, and likes repel**.

1.1 The Stator, The Rotor, and Magnetic Poles

Every brushless motor consists of two mechanical sub-assemblies:
• **The Stator (Outer Stationary Ring):** Made of laminated silicon steel teeth wrapped with copper wire coils grouped into three phases: **Phase A, Phase B, and Phase C**.
• **The Rotor (Inner Spinning Shaft):** A cylindrical shaft carrying powerful permanent magnets (Neodymium-Iron-Boron) with alternating North (N) and South (S) magnetic poles.
When electric current flows through a copper coil, it creates an electromagnet (Ampère's Law). By feeding alternating currents into Phases A, B, and C at slightly different times, we create a smooth, rotating magnetic field that sweeps around the stator ring.

> **LAYMAN'S ANALOGY**
>
> ***The Spinning Donut Analogy:*** *Imagine holding a compass needle in your hand. If you hold a strong magnet near the compass and walk in a circle around it, the needle will pivot and chase the magnet continuously. In an electric motor, the stator coils act as the circling magnet, and the rotor's permanent magnets act as the needle! By commanding the copper coils to pull in a continuous circle, we drag the shaft along at thousands of RPM.*

1.2 The 'Carrot on a Stick' Principle: 90° Torque Angle

Here is the golden rule of electric motor efficiency: **To get maximum torque (twisting force) from every watt of battery power, the stator's magnetic field must lead the rotor's permanent magnets by exactly 90 electrical degrees.**
• If the stator field pulls directly in line with the rotor magnet (0° angle), it simply pulls the rotor outwards against the motor bearings. It generates **zero torque** and turns 100% of your battery power into wasted heat!
• If the stator field pulls at 90°, all magnetic attraction acts tangentially, twisting the shaft with maximum possible leverage.
• Like a carrot dangling in front of a donkey: if the carrot is right in front of its nose, it chases it with maximum enthusiasm. If you dangle it behind its head, it stops!

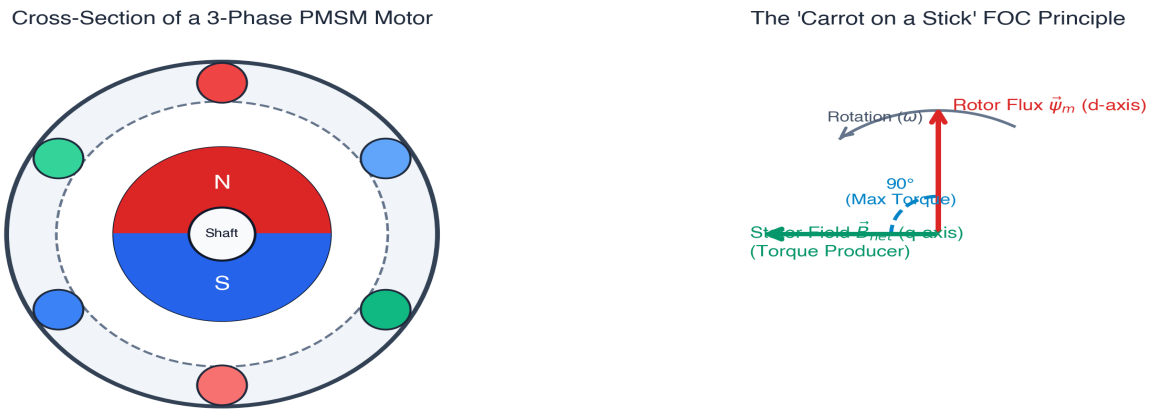


Figure 1: (Left) Cross-section of a 3-Phase PMSM with stator coils and rotor. (Right) The 90° torque vector angle.

1.3 Back-EMF & Faraday's Law: The Motor as a Generator

Whenever a permanent magnet sweeps past a copper coil, it pushes electrons through the wire, generating a voltage. This is **Faraday's Law of Electromagnetic Induction**. While our motor is spinning, it is simultaneously acting as an electrical generator! The voltage it generates pushes backward against the battery voltage — which is why it is called **Back-Electromotive Force (Back-EMF)**.

The Secret of Sensorless Control: Back-EMF is not a nuisance — it is a treasure trove of information! Its amplitude tells us exactly how fast the motor is spinning, and its phase tells us exactly where the rotor is pointing. By reading the motor's voltages and currents, our software can mathematically deduce the Back-EMF and 'see' the rotor angle without physical sensors!

1.4 The Standstill Blind Spot & Open-Loop Startup (I-F Ramp)

Because Back-EMF depends entirely on magnet motion, when the motor is stopped (0 RPM), **the Back-EMF is exactly zero volts**. At standstill, the motor is completely blind — the observer cannot see where the rotor is because no voltage is being induced. To overcome this, our firmware uses a robust 3-stage startup sequence:

1. **ALIGN (Mode 0):** We inject a steady DC current into Phase A for 500 ms. This pulls the rotor into alignment with Phase A, locking it to a known 0° position.
2. **OPEN_LOOP_RAMP (Mode 1):** We synthesize a rotating current vector that smoothly accelerates from 0 to 500 RPM (I-F ramp), dragging the rotor into motion.
3. **CLOSED_LOOP Handoff (Mode 2):** At 500 RPM, Back-EMF is large and clean. The observer locks on; once the observer angle matches the forced angle within 15°, full control is seamlessly handed over to the sensorless algorithm!

Chapter 2: Field-Oriented Control (FOC) Demystified

In a conventional 3-phase motor, currents in wire A, wire B, and wire C are alternating current (AC) sine waves oscillating 120° apart. Controlling three oscillating waves simultaneously is like juggling spinning plates while riding a unicycle. Field-Oriented Control (FOC) solves this by using two coordinate transforms that turn oscillating AC signals into flat DC values.

2.1 The Clarke Transform: 3 Phases to 2 Orthogonal Axes

The Clarke Transform converts the 3 physical phase currents (i_a, i_b, i_c) situated 120° apart into two stationary orthogonal axes situated 90° apart: **Alpha (α) and Beta (β)**:

$i_\alpha = i_a$

$i_\beta = \frac{1}{\sqrt{3}} (i_a + 2 i_b)$

This reduces the mathematical dimensionality from 3 down to 2 while preserving 100% of the electromagnetic information.

2.2 The Park Transform: Riding the Spinning Rotor Frame

While i_α and i_β are simpler, they are still rapidly oscillating AC waves. The Park Transform performs an act of mathematical genius: it rotates the reference frame at the exact speed and angle θ of the spinning rotor!

$i_d = i_\alpha \cos\theta + i_\beta \sin\theta$ (Direct axis, flux alignment)

$i_q = -i_\alpha \sin\theta + i_\beta \cos\theta$ (Quadrature axis, torque producer)

Think of riding a carousel: to someone standing on the ground, you are spinning in circles. But to someone sitting next to you on the carousel, you appear completely stationary! In the (d, q) frame, **the AC currents turn into steady DC numbers**. Controlling motor torque becomes as simple as controlling a basic brushed DC motor: set i_q to your desired torque, and regulate $i_d = 0$!

Field-Oriented Control (FOC) Closed-Loop Mathematical Architecture

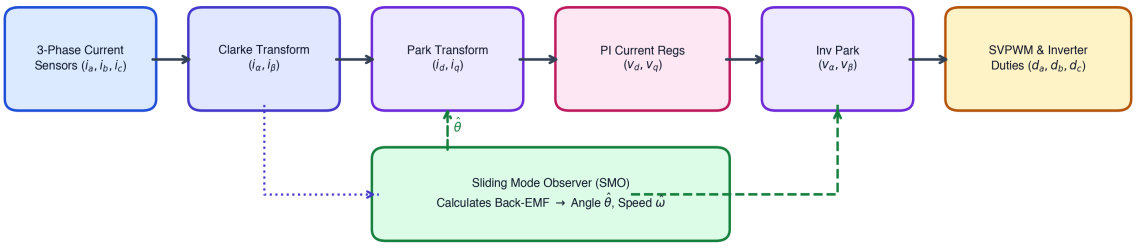


Figure 2: The complete closed-loop Field-Oriented Control (FOC) pipeline with Sliding Mode Observer feedback.

2.3 PI Controllers and Space Vector PWM (SVPWM)

Two Proportional-Integral (PI) controllers compare measured (i_d, i_q) against target setpoints and compute required voltages (v_d, v_q). The **Inverse Park Transform** rotates these back into stationary (α, β) coordinates. Finally, **Space Vector Pulse-Width Modulation (SVPWM)** calculates the switching duty cycles (d_a, d_b, d_c) to drive the inverter MOSFETs at 20 kHz. SVPWM uses the DC bus 15.5% more efficiently than sinusoidal PWM, producing higher top speeds and maximum battery efficiency.

Chapter 3: The Virtual Sensor — Sliding Mode & Flux Observers

A physical sensor measures reality with hardware; a **State Observer** deduces reality through mathematical models. Our controller runs an exact digital twin of the motor's electrical circuits inside the ESP32 CPU 20,000 times per second.

3.1 The Sliding Mode Observer (SMO)

The observer models stator coil differential equations: $\frac{d\hat{i}_\alpha}{dt} = \frac{1}{L_s}(v_\alpha - R_s \hat{i}_\alpha - \hat{e}_\alpha)$ and $\frac{d\hat{i}_\beta}{dt} = \frac{1}{L_s}(v_\beta - R_s \hat{i}_\beta - \hat{e}_\beta)$.
At every 50 µs tick, the code compares measured currents with modeled currents. If there is an error, the Sliding Mode Observer uses a switching correction function to slide the model until the current error is driven to zero. The correction signal required to force the model to match reality **is the physical Back-EMF $(\hat{e}_\alpha, \hat{e}_\beta)$**! Taking $-\text{atan2}(\hat{e}_\alpha, \hat{e}_\beta)$ yields the rotor angle $\hat{\theta}$, which is smoothed by a Phase-Locked Loop (PLL) to extract rotor speed $\hat{\omega}$.

OBSERVER ARCHITECTURE

Dual-Observer Architecture in this Repo:
- ***Sliding Mode Observer (SMO):*** *Ultra-fast dynamic response; tracks aggressive speed steps and load shocks effortlessly.*
- ***Flux Observer:*** *Integrates stator flux vectors; provides silky smooth estimation during steady-state cruising.*
- ***Observer Manager:*** *Cross-checks both observers simultaneously. If one diverges or reports inconsistent flux magnitude, it flags an internal tracking error and alerts the fault isolation supervisor.*

Chapter 4: Inverter Realities — Dead-Time, Body Diodes, & Bus Dynamics

A physical inverter consists of 6 power MOSFETs arranged into 3 half-bridges. Hardware engineering introduces several physical non-idealities:

4.1 Shoot-Through and Dead-Time Distortion

If the high-side and low-side MOSFETs of a half-bridge turn on at the same time, battery voltage is shorted directly to Ground (**shoot-through**), destroying the transistors in nanoseconds! To prevent this, gate drivers enforce an intentional **dead-time** ($t_{dead} \approx 1.0\text{ µs}$) where both switches are off. During dead-time, phase current is forced through the MOSFET parasitic **body diodes**, creating a voltage distortion ($V_{dead} = t_{dead} \cdot f_{pwm} \cdot V_{bus}$). Our controller incorporates dead-time compensation, but on uncommissioned hardware defaults to 0.0 s to avoid overcompensation (which would introduce up to 19.3° angle error).

4.2 Regenerative Braking & Safe Stop Diode Conduction

When the controller commands **SAFE_STOP (Mode 5)**, all 6 MOSFETs are disabled (`PIN_GATE_EN = 0`). The spinning motor acts as a 3-phase generator. If its Back-EMF exceeds the 24 V bus, current rectifies through the MOSFET body diodes into the DC bus capacitor, causing bus voltage to rise slightly (e.g. from 24.0 V to 24.45 V). A physical bleeder resistor ($R_{bleed} = 50\text{ }\Omega$) safely dissipates this energy, restoring nominal voltage within ~100 ms.

Chapter 5: How Faults Are Detected & Isolated (The Core Innovation)

In critical applications like drones and robotics, stopping mid-motion because of a sensor failure can be fatal. Our controller provides fail-operational fault tolerance through three physical tiers:

5.1 Kirchhoff's Current Law (KCL): The Universal Truth

In a 3-phase star-connected motor, electrical charge cannot vanish. By **Kirchhoff's Current Law**, the instantaneous sum of all three currents must always equal zero:

$i_a(t) + i_b(t) + i_c(t) = 0$

Our code computes the KCL Residual: **$r_{KCL} = |i_a + i_b + i_c|$**. In healthy operation, $r_{KCL} < 0.15$ A. If any sensor develops a bias or gets stuck, r_{KCL} leaps above our **0.6 A** threshold!

5.2 Hypothesis Testing & 2-Shunt Degraded Handoff

KCL tells us a sensor is broken, but not *which* one! The algorithm evaluates three hypotheses against the observer model:

- **Hypothesis A:** Sensor A is broken $\to$ Reconstruct $i_a = -(i_b + i_c)$. Check model error E_a.
- **Hypothesis B:** Sensor B is broken $\to$ Reconstruct $i_b = -(i_a + i_c)$. Check model error E_b.
- **Hypothesis C:** Sensor C is broken $\to$ Reconstruct $i_c = -(i_a + i_b)$. Check model error E_c.

The true healthy pair matches the digital twin model with tiny error; the broken sensor creates massive discrepancy. Within 15 ms, the bad sensor is isolated and discarded, and the system transitions to **DEGRADED Mode (Mode 3)**, continuing to run on 2 shunts!

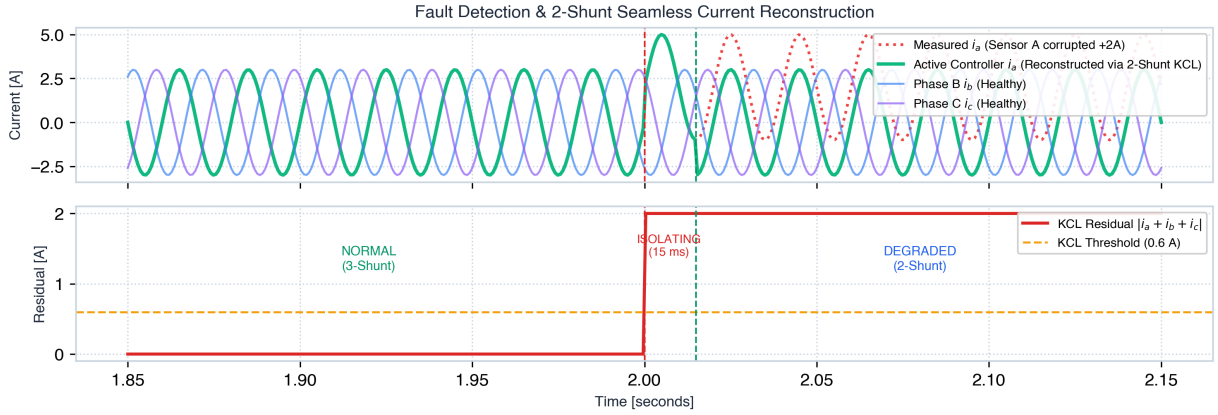


Figure 3: (Top) Seamless transition to 2-shunt current reconstruction upon Sensor A +2A bias. (Bottom) KCL residual tripping isolation.

5.3 Debounce Envelope & Noise Immunity

Power electronics generate severe electrical switching noise. A stray EMI glitch might cause one sample to read +3 A for 50 µs. Our multi-tier debounce filter prevents false trips:

- **1–2 samples up to ± 5.0 A:** Completely rejected! Mode stays `CLOSED_LOOP`, speed dip < 3%.
- **4 samples at 1.4 A:** Filtered out (does not meet 5-sample consecutive limit).
- **5 samples at 0.5 A:** Filtered out (within noise margin).
- **≥ 3 consecutive samples ≥ 1.5 A (or ≥ 5 samples ≥ 0.6 A):** Confirms a true physical fault and latches isolation within 150 µs!

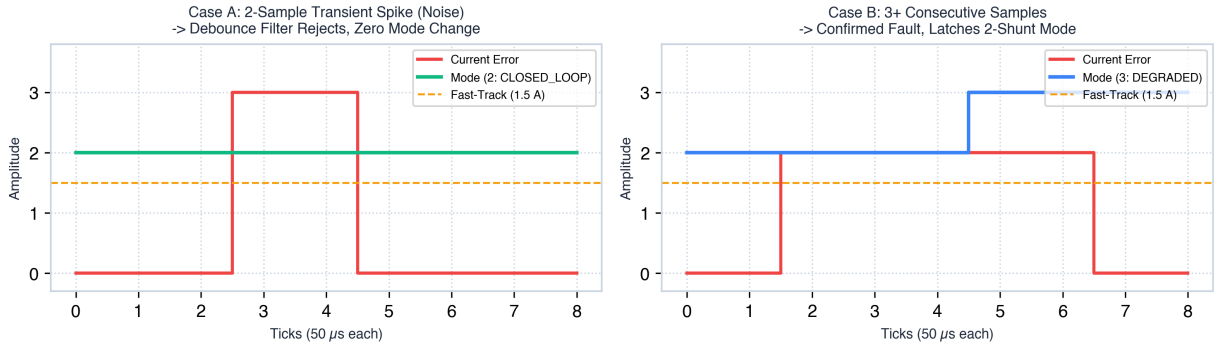


Figure 4: Debounce envelope rejecting short transient EMI spikes (Left) while rapidly latching sustained faults (Right).

Chapter 6: Detailed Breakdown of All 13 Test Scenarios

Every scenario in `simulation/fault_scenarios.yaml` tests a distinct physical failure mode in high-fidelity Software-in-the-Loop simulation:

#	Scenario Name	Physical Fault Injected	Acceptance Criteria & Behavior	Result
1	**Baseline Healthy**	No faults. Smooth accel to 2000 RPM.	Angle error < 15°, speed ripple < 2%, health = 100%.	**PASS**
2	**Phase A +2A Offset**	+2.0 A current bias on Sensor A at t=2.0s.	Detect < 15 ms, isolate A, stay in DEGRADED, error < 30°.	**PASS**
3	**Phase B +2A Offset**	+2.0 A current bias on Sensor B at t=2.0s.	Detect < 15 ms, isolate B, stay in DEGRADED, error < 30°.	**PASS**
4	**Phase C +2A Offset**	+2.0 A current bias on Sensor C at t=2.0s.	Detect < 15 ms, isolate C, stay in DEGRADED, error < 30°.	**PASS**
5	**Sensor A Stuck Zero**	Sensor A output freezes at 0.0 A at t=2.0s.	Detect zero-variance, isolate A, 2-shunt handoff.	**PASS**
6	**Inverter Leg A Open**	Phase A inverter leg disconnects (0 current).	Severe residual, latch SAFE_STOP within 50 ms.	**PASS**
7	**Handoff / Alignment**	Standstill DC align to 500 RPM I-F ramp.	Observer locks < 15°, handoff to CLOSED_LOOP clean.	**PASS**
8	**Parameter Mismatch**	Motor R_s, L_s detuned by $\pm 20\%$.	Observer maintains lock, angle error < 25°, stable.	**PASS**
9	**Speed Reversal**	Speed step from +2000 to -2000 RPM.	Smooth zero-speed crossing, 4-quadrant stability.	**PASS**
10	**Torque Step (+0.5Nm)**	Sudden mechanical load applied at t=2.5s.	Speed dip < 5%, recovery < 80 ms, PI anti-windup.	**PASS**
11	**Two-Phase Startup**	Sensor A disconnected before startup.	Starts directly in 2-shunt mode, reaches 2000 RPM.	**PASS**
12	**Intermittent Fault**	Phase A sensor chatters on/off every 50 ms.	Isolates to 2-shunt, debounce prevents flapping.	**PASS**
13	**Noise Spike Immunity**	+3.0 A single-sample EMI spikes injected.	Mode stays CLOSED_LOOP (2), speed dip < 3%.	**PASS**

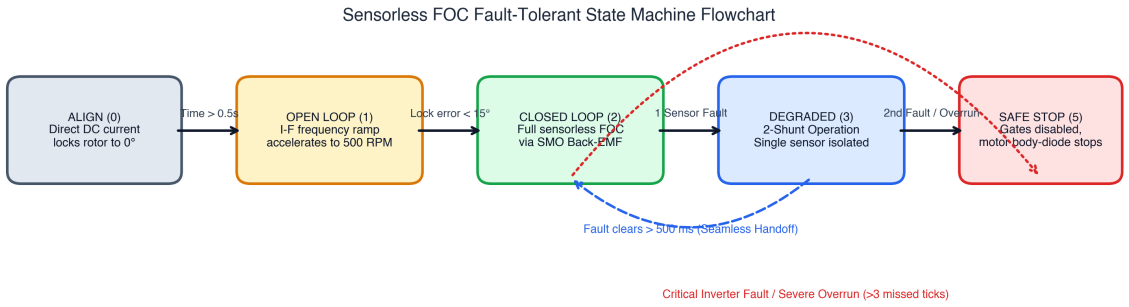


Figure 5: Sensorless FOC Fault-Tolerant State Machine transition flow between operational modes.

Chapter 7: ESP32 Microcontroller Implementation & Real-Time HAL

Porting vector FOC algorithms to an embedded dual-core MCU (ESP32 @ 240 MHz) requires strict real-time guarantees:

• **Core 0 vs Core 1 Partitioning:** Core 0 runs low-priority telemetry, Serial printing, and user I/O. Core 1 runs the 20 kHz FOC task exclusively with zero RTOS lock contention.

• **Ultra-Lean 20 kHz Timer ISR:** Uses `vTaskNotifyGiveFromISR()` (< 1 µs overhead) to wake the pinned Core 1 control task cleanly without running heavy math in interrupt context.

• **Direct Hardware Register SAR ADC1 Reads:** Bypasses FreeRTOS mutexes to poll `SENS.sar_meas_start1` directly with a

bounded **15 µs conversion timeout** and immediate error bailout.
• **Real-Time Overrun Protection:** Uses `ulTaskNotifyTake` return value to detect missed deadlines; 3 consecutive missed ticks trips `SAFE_STOP`.
• **CPU Cycle-Counter Benchmarking (`-DBENCH_TIMING`):** Uses `xthal_get_ccount` (4.167 ns resolution) to profile ADC read time, control step time, total task time, and wake jitter without an oscilloscope.

Embedded Metric	Specification / Measured Value	Verification Method
PWM Switching Freq	20,000 Hz (50.0 µs period)	Hardware Timer 0 Alarm at 50 µs
ADC Sampling Channels	3 channels (GPIO 32, 33, 34 / ADC1_CH4,5,6)	Fast register SAR polling via SENS struct
ADC Timeout Limit	15.0 µs per channel (abort < 50 µs tick)	Bounded hardware timer loop in HAL
Static RAM Usage	22,212 bytes (6.8% of 320 KB)	Cross-compiled ELF size via pio run
Flash Memory Usage	299,545 bytes (22.9% of 1.3 MB)	Flash footprint verified in pio_build.log
Overrun Protection	3 consecutive missed ticks trips SAFE_STOP	ulTaskNotifyTake return value tracking
Physical Probe Pin	GPIO 14 (PIN_TIMING_BENCHMARK)	HIGH during task execution for scope probe

Chapter 8: Glossary of Technical Terms for Non-Experts

Term	Plain-English Definition & Function
FOC (Field-Oriented Control)	A high-performance control method that coordinates AC motor currents into two constant DC components: one for magnetizing flux (i_d) and one for pure mechanical torque (i_q).
PMSM	Permanent Magnet Synchronous Motor — an AC motor whose spinning rotor contains permanent magnets and rotates in exact synchronism with the stator's rotating magnetic field.
BLDC	Brushless Direct Current Motor — similar to PMSM, but typically wound for trapezoidal Back-EMF and driven by simpler 6-step block commutation.
Back-EMF	Back-Electromotive Force: the voltage naturally generated inside motor coils when permanent magnets spin past them. Used as a 'virtual sensor' to deduce rotor angle.
Clarke Transform	A mathematical formula that converts 3-phase AC currents (i_a, i_b, i_c) into two stationary orthogonal coordinates (i_α, i_β).
Park Transform	A mathematical formula that rotates coordinates along with the rotor, transforming oscillating (α, β) signals into flat DC (d, q) values.
SVPWM	Space Vector Pulse-Width Modulation — an advanced technique for switching inverter MOSFETs that extracts 15.5% more voltage from a battery than standard sinusoidal PWM.
SMO (Sliding Mode Observer)	A robust mathematical observer that forces a software model of the motor to track real currents, exposing the Back-EMF voltages needed for sensorless angle tracking.
KCL (Kirchhoff's Current Law)	A fundamental law of physics stating that the sum of all currents entering a node must equal zero. In a motor: $i_a + i_b + i_c = 0$.
Dead-Time	A brief microsecond pause inserted between turning off one MOSFET and turning on another in the same leg to prevent a dead-short across the power supply (shoot-through).
2-Shunt Degraded Mode	A fault-tolerant mode where one broken current sensor is discarded and its current is reconstructed mathematically from the remaining two healthy sensors ($i_a = -(i_b + i_c)$).

ENGINEERING CONCLUSION

Conclusion & Engineering Takeaway: *By marrying high-speed mathematical transformations (Clarke/Park), digital twin observers (SMO/Flux), physical power electronics protections (dead-time compensation, body diode conduction), and robust Kirchhoff-based hypothesis testing, this repository demonstrates that low-cost microcontrollers ($<\$5$) like the ESP32 can deliver mission-critical, fail-operational motor control previously reserved for aerospace hardware.*